التاريخ: / /

موضوع الدرس

بسم الله الرحمن الرحيم.

section (7) Deflection Magnetometer.

1). Name of theory → . Deflection Magnetometer.

2). Aim of theory → . to compare between the magentic ~~momen~~ dipole moment of 2 bar magnets.

3). used law → $\frac{M_1}{M_2} = \frac{\tan\theta_1}{\tan\theta_2}$.

4). theory → 1). magnetic flux → . Is the amount of magnetic field lines passing through a given surface.

2). Magnetic field → . is a region around amagnetic material or amoving electric charge where magnetic forces can be observed.

3). Magnetic Moment العزم → is a vector quantity that represents the strength and direction of a magnetic force.

4). Gauss's law → states that the net magnetic flux through any closed surface is always zero.

5). Resultant محصلة of 2 perpendicular متعامدين field → if 2 magnetic fields B_1, B_2 are perpendicular, then the resultant field B is → $B = \sqrt{B_1^2 + B_2^2}$.

5). tools → 2 bar magnets - Magnetic needle "إبرة" بوصلة compass ruler.

6). steps → 1). Place the magnetic needle on a flat surface.

2). set the first magnet at a fixed distance from the needle.

3). measure and record the angle of deflection of first magnet.

4). Repeat the measurement 8 times.

5). Replace the first magnet with the second one, keeping the same distance. do measurement for second magnet 8 times, "the angle of deflection"

6). Compare the magnetic moments based on the average deflection angles.

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7). Results: magnet 1: $\theta_1 = 49$, $\theta_2 = 51$, $\theta_3 = 40$, $\theta_4 = 39$, $\theta_5 = 45$, $\theta_6 = 46$, $\theta_7 = 50$, $\theta_8 = 51$.

magnet 2: $\theta_1 = 47$, $\theta_2 = 49$, $\theta_3 = 41$, $\theta_4 = 43$, $\theta_5 = 49$, $\theta_6 = 51$, $\theta_7 = 38$, $\theta_8 = 39$.

average $\theta_1 = \frac{\theta_1 + \theta_2 + \theta_3 + \theta_4 + \theta_5 + \theta_6 + \theta_7 + \theta_8}{8} = 46.375$

average $\theta_2 = \frac{\theta_1 + \theta_2 + \theta_3 + \theta_4 + \theta_5 + \theta_6 + \theta_7 + \theta_8}{8} = 44.625$

$\therefore \frac{M_1}{M_2} = \frac{\tan\theta_1}{\tan\theta_2} = \frac{\tan 46.375}{\tan 44.625} = 1.06$.